Vijay Mohanram Iyer

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SUMMARY

Graduate engineer with 3 years of experience specializing in intelligent safety systems and health applications powered by computer vision and machine learning. Skilled in transforming advanced algorithms including deep learning architectures and large language models into practical, user-centered tools that enhance security and promote human well-being.

PROFESSIONAL EXPERIENCE

1. Forschungszentrum Informatik — *September 2025 – Present*
   Research Assistant
   - Evaluate and optmize detection models through transfer learning and ensemble methods for hyperparameter fine-tuning and multimodal data analysis.
   - Exploring GAN based approaches and Latent-Diffusion techniques and their architectures for forged video rendering .
   - To utilize 3D geometric facemesh for reconstruction of desired regions.
   - Conduct literature review and work towards research publication.

2. TecoLab — *March 2023 – Present*
   Working Student
   - Sensor optimization for open-earables with TinyML and quantized neural networks for on-device inference.
   - ML4Print: Classification based on printer-specific characteristics and paper substrate type using feature extraction and supervised ML classifiers.
   - Computational Fluid Dynamics: utilized regression models and data integration to simulate the thermal behavior of liquids within industrial valves, thereby enhancing predictive maintenance capabilities and system reliability.

3. Forschungszentrum Informatik — *February 2025 – August 2025*
   Master Thesis (Grade: 1,0)
   - Developed an end-to-end deep learning pipeline using rPPG signals for artifact evaluation in vital signals and Benchmarked under real-world conditions.
   - Proposed a Fusion Model combining Vision Transformer and CNN features via a custom Fusion-Head for DeepFake classification.

4. Access@KIT — *March 2023 – May 2023*
   Working Student
   - Developed a bi-directional LSTM-based text-to-speech system with phoneme alignment and acoustic feature extraction, optimizing temporal dependencies and prosody for accessible, natural speech for visually impaired users.

5. Accenture India Pvt. Ltd. *February 2022 – April 2022*
Associate Software Engineer

- Maintained IBM Mainframe servers with batch job scheduling and performance monitoring COBOL scripts.

EDUCATION

M.Sc. in Electrical Engineering and Information Technology *May 2022 – July 2025*
Karlsruhe Institute of Technology
GPA: 2.3

Bachelor of Engineering in Electronics Engineering *August 2017 – May 2021*
University of Mumbai, India
GPA: 2.8

PROJECTS

1. **Attention Mechanism and Multi-ROI Artifact Recognition for DeepFake Detection Using rPPG (Forschungszentrum Informatik - Thesis)** *February 2025 – July 2025*
Utilized POS and CWT to extract pulsatile components for detecting signal irregularities in real vs. forged videos, enhancing DeepFake detection for privacy and security.
Technology: Python, PyTorch, OpenCV, NumPy, Matplotlib, ViT, rPPG, GenAI
2. **CamCussion: Eye Tracking Software (Zeiss Innovation Hub)** *November 2023 – July 2024*
Utilized pupil segmentation, gaze tracking, and saccadic velocity estimation to analyze pupil dilation and eye movements in real time for accurate concussion diagnosis
Technology: Python, OpenCV, Dlib, PyQt, image processing, PyTorch, NumPy, ML
3. **Self-Driving Car using LIDAR** *September 2022 – February 2023*
Developed real-time 3D obstacle detection by voxelizing LiDAR point clouds into ViT tokens and applying Multi-head attention for improved accuracy.
Technology: Python, PyTorch, Open3D, NumPy, ViT, LiDAR processing
4. **Real-Time Car Accident Alert System** *January 2022 – June 2022*
Built ESP32-based embedded LSTM system with TensorFlow Lite for real-time crash detection and automated SMS/email alerts with GPS location.
Technology: Python, TensorFlow Lite, ESP32, LSTM, Embedded C

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL, Node.js, HTML, CSS
- **Libraries/Frameworks:** OpenCV, Qt, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, ROS, React.js
- **ADAS:** Sensor fusion, camera and LiDAR calibration, perception algorithms, path planning, object detection & tracking
- **Tools & Cloud Platforms:** Git, Docker, MATLAB, Linux, Carla, conda, LaTeX, Jupyter Notebook, Azure, AWS
- **Spoken Languages:** English C1, German A2 (Learning)

CORE COMPETENCIES

- **Technical Leadership:** Project Development, Cross-functional Collaboration, Innovative mindset.
- **Problem Solving:** Critical Thinking, Performance Optimization, Debugging.
- **Communication:** Active Listening, Documenting, Knowledge Transfer.